

CASE STUDY

Library Management Database

Designing a 3NF relational database in MS-SQL Server

Project	Library Management & Fine Tracking Database
Database	LibraryManagement
Platform	Microsoft SQL Server (T-SQL)
Design Level	Third Normal Form (3NF)
Tables	6 (5 Master, 1 Detail)
Program	IsDB-BISEW – Project: Database Creation in MS-SQL
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1 Project Name

Project: Library Management & Fine Tracking Database.

2 Background

Situation: In a library, members borrow books for a fixed period (10 days in the sample data). For every borrowing, the library records who took which book, the issue date, the return deadline, the actual return date and, for late returns, the fine.

Problem: When this is kept in one flat record, member, book and author details repeat again and again. It becomes hard to find overdue books, track member activity or calculate fines correctly.

Solution: I organized this data into a relational database to manage the library records and to make analysis easy.

3 Objective

Goal: To transform the flat library record into a structured 3NF relational database that removes data redundancy and allows easy querying of issued and returned books, late returns, member activity, most-issued books and authors, and fine calculation based on the fine policy.

4 Design: Entities and their Roles

Created a 3NF database based on the following six entities:

Entity	Type	Role Description
Author	Master	Master list of authors of the books in the library.
Category	Master	Master list of book categories (e.g., History, Poetry, Novel).
Member	Master	Stores member information: name, city, membership date and membership expiry date.
FinePolicy	Master	Defines the fine rules: fine type (e.g., No fine, Per Day) and the fine amount.
Book	Master	Stores book information: book name, its author and its category.
BookIssue	Detail	Book issue transactions: which member took which book, issue date, return deadline, return date and the fine policy applied.

5 Sample Table

Member Name	Book Name	Author Name	Category Name	Issue Date	Return Dead Line	Return Date	Fine Amount	Fine Type
Al Amin	Les Miserables	Victor Hugo	History	2025-10-10	2025-10-20	2025-10-19	0.00	No fine
Monir Hossain	Chayanat	Kazi Nazrul Islam	Poetry	2025-11-05	2025-11-15	2025-11-25	20.00	Per Day
Sajid Hossain	Animal Farm	George Orwell	Novel	2025-12-02	2025-12-12	2025-12-11	0.00	No fine
Abdul Kader	Gora	Rabindranath Tagore	Novel	2026-01-10	2026-01-20	2026-01-19	0.00	No fine
Shakil Ahmed	Bidrohi	Kazi Nazrul Islam	Poetry	2026-02-05	2026-02-15	2026-02-14	0.00	No fine
Tanvir Ahmed	Gitanjali	Rabindranath Tagore	Poetry	2026-03-02	2026-03-12	2026-03-11	0.00	No fine
Arif Hossain	Pather Dabi	Sarat Chandra Chattopadhyay	Novel	2026-03-05	2026-03-15	2026-03-22	20.00	Per Day
Shakil Ahmed	Beder Meye	JasimUddin	Novel	2026-03-09	2026-03-19	2026-03-18	0.00	No fine
Rakib Hossain	Chander Pahar	Shirshendu Mukhopadhyay	Novel	2026-03-16	2026-03-26	2026-04-02	20.00	Per Day
Mamun Mia	Sei Somoy	Sunil Gangopadhyay	History	2026-04-05	2026-04-15	2026-04-18	10.00	Per Day
Shakil Ahmed	Hamlet	William Shakespeare	Novel	2026-04-12	2026-04-22	2026-04-21	0.00	No fine

Highlighted rows are late returns (Return Date after Return Dead Line) with a fine.

6 How They Are Connected

Relationship	Type	Linked By	Purpose
Author → Book	One-to-Many	AuthorID	Each book is written by one author; one author can have many books.
Category → Book	One-to-Many	CategoryID	Each book belongs to one category; one category can have many books.
Member → BookIssue	One-to-Many	MemberID	Every issue record belongs to a member; a member can issue many books.
Book → BookIssue	One-to-Many	BookID	Every issue record is for one book; a book can be issued many times.
FinePolicy → BookIssue	One-to-Many	FinePolicyID	The fine rule applied to an issue record, used to calculate the fine for late returns.

Note: Member and Book have a many-to-many relationship, which is resolved through the BookIssue table.

7 ER Diagram

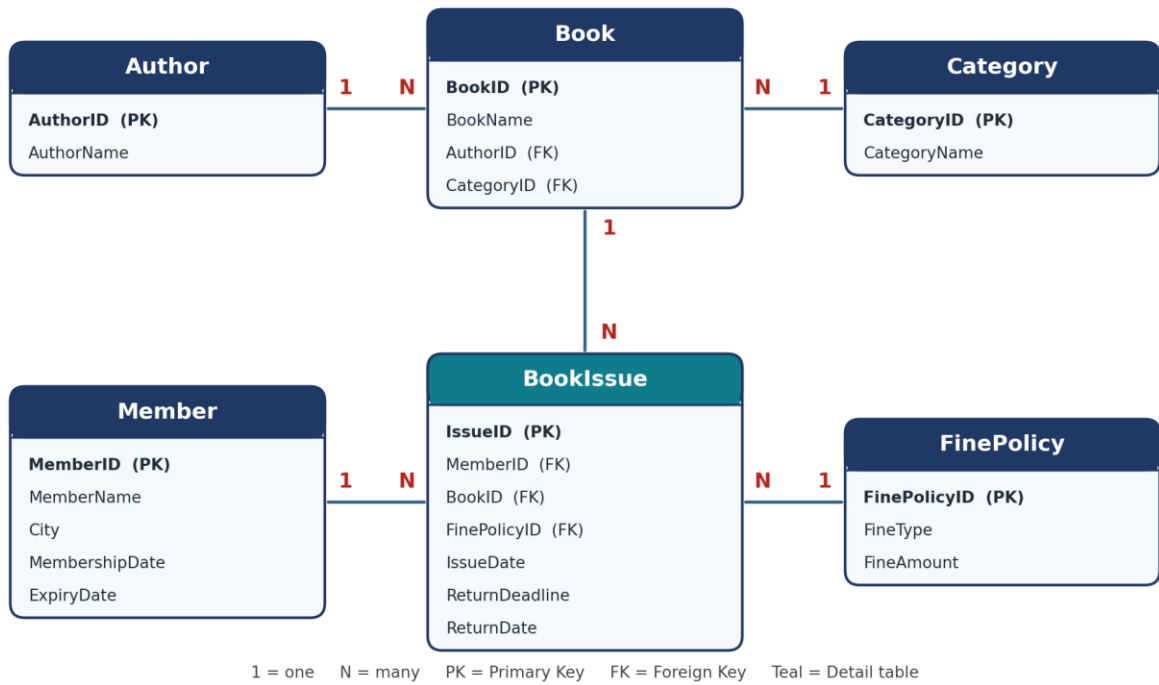


Figure: Entity Relationship Diagram of the Library Management Database

Section and Serial-wise Queries

- 1. DDL-1.** Create a 3NF database and tables by writing sql script as follows: Database Name: LibraryManagement, Data File Name: LibraryManagement_Data_1, Log File Name: LibraryManagement_Log_1, Location: default database file location, Size (data file: 30mb, log file: 2mb), Maximum size (data file: 100mb, log file: 50mb), File growth (data file: 5%, log file: 1mb)
- 2. DDL-2.** Creating Tables for the Database.
- 3. DML-1.** Data insertion into Tables.
- 4. DML-2.** Write a query to find all members who currently have books that are not yet returned.
- 5. DML-3.** Write a query to retrieve the list of books that have been issued 2 times or more.
- 6. DML-4.** Write a query to find members who have returned books after the deadline.
- 7. DML-5.** Write a query to calculate fine amount collected based on FinePolicy for all returned books.
- 8. DML-6.** Write a query to find the most issued books in the library.
- 9. DML-7.** Write a query to retrieve members whose membership has already expired but still have issued books.
- 10. DML-8.** Write a query to find members who have never issued any book.
- 11. DML-9.** Write a query to retrieve the top 3 members who issued the highest number of books.
- 12. DML-10.** Write a query to retrieve all book issue transactions where the book has been returned, and either the issue date is later than January 1, 2025 or the assigned return deadline exceeds June 1, 2025.
- 13. DML-11.** Write a query to retrieve members whose city are all except 'Dhaka', 'Chattogram', 'Cumilla', and membership date is later than '2025-06-01'.

- 14. DML-12.** Write a query to retrieve book issue records from '2025-01-01' to '2025-03-31'.
- 15. DML-13.** Write a query to find all members whose city starts with 'D' and the next letter is one of A through M.
- 16. DML-14.** Write a query to find all members whose city starts with 'D' and the next letter is not in N through Z.
- 17. DML-15.** Write a query to retrieve 7 through 15 records from BookIssue table.
- 18. DML-16.** Write a query to find members who have never returned any book late.
- 19. DML-17.** Write a query to find authors whose books have never been issued.
- 20. DML-18.** Update Row
- 21. DML-19.** Delete Row
- 22. DML-20.** Use of Aggregate function.
- 23. DML-21.** Example of Using Merge Between two tables.
- 24. DML-22.** Example of Using CUBE Operator.
- 25. DML-23.** Example of Using Rollup Operator.
- 26. DML-24.** Example of Using Grouping Sets Operator.
- 27. DML-25.** Use of Over Clause.
- 28. DML-26.** Example of Data Sorting.
- 29. DML-27.** Example of Using Subquery.
- 30. DML-28.** Use of Logical Operator-
 - a. All keyword.
 - b. Any/some keyword.
- 31. DML-29.** Example of Using Correlated Subquery.
- 32. DML-30.** Use of Exists Operator.
- 33. DML-31.** Example of using Common Table Expressions (CTE).
- 34. DML-32.** Use of Case Function.
- 35. DML-33.** Use of Search Case Function.
- 36. DML-34.** Use of IIF Function.
- 37. DML-35.** Example of Using NULL Handling functions (COALESCE/ISNULL).

38. DML-36. Use of Ranking Functions

- a. ROW_NUMBER,
- b. RANK,
- c. DENSE_RANK,
- d. NTILE

39. DML-37. Example of converting Data type-

- a. Cast,
- b. Convert,
- c. Try_Convert

40. DML-38. Use of Outer Join-

- a. Left join,
- b. Right Join

41. DML-39. Example of Using Set Operator (Union operator).

42. DML-40. Use of analytical functions-

- a. First_Value,
- b. Last_Value,
- c. LEAD,
- d. LAG,
- e. Cumulative Distribution (Cume_Dist)

43. DML-41. Use of @@identity And @@Rowcount (WITH TRANSACTION)

44. DML-42. Justification of Using @@identity And @@Rowcount.

45. DDL-3. Example of Creating View.

46. DML-43. Justification of View

47. DDL-4. Example of Creating View with Encryption.

48. DML-44. Justification of view with Encryption

- 49. **DDL-5.** Example of Creating View with Schemabinding.
- 50. **DML-45.** Justification of View with Schemabinding
- 51. **DDL-6.** Example of Creating View with encryption and schemabinding Together.
- 52. **DML-46.** Justification of View with encryption and schemabinding
- 53. **DDL-7.** Example of View Modification (**Alter View**).
- 54. **DML-47.** Justification of Alter View
- 55. **DDL-8.** Example of Updatable View.
- 56. **DML-48.** Justification of Updatable View
- 57. **DDL-9.** Delete View.
- 58. **DDL-10.** Example of Scalar function.
- 59. **DML-49.** Justification of Scalar function
- 60. **DDL-11.** Example of Simple Table valued function.
- 61. **DML-50.** Justification of Simple Table valued function
- 62. **DDL-12.** Example of Multi Statement Table valued function.
- 63. **DML-51.** Justification of Multi Statement Table valued function
- 64. **DDL-13.** Function with encryption and schemabinding Altogether.
- 65. **DML-52.** Justification of function with encryption and schemabinding
- 66. **DDL-14.** Example of Function Modification (Alter function).
- 67. **DML-53.** Justification of **ALTER FUNCTION**.
- 68. **DDL-15.** DELETE FUNCTION.
- 69. **DDL-16.** Stored Procedure (Without Parameter).
- 70. **DML-54.** Justification of STORED PROCEDURE(Without Parameter)
- 71. **DDL-17.** Stored Procedure (With Parameter).
- 72. **DML-55.** Justification of STORED PROCEDURE(With Parameter)
- 73. **DDL-18.** Stored Procedure (With optional Parameter).
- 74. **DML-56.** Justification of STORED PROCEDURE(With optional Parameter)
- 75. **DDL-19.** Procedure with Recompile.
- 76. **DML-57.** Justification of Procedure with recompile
- 77. **DDL-20.** Procedure with Encryption.

- 78. DML-58.** Justification of procedure with encryption
- 79. DDL-21.** Example of Data insertion through Procedure.
- 80. DML-59.** Justification of Data insertion through Procedure
- 81. DDL-22.** Example of Data updating through Procedure.
- 82. DML-60.** Justification of Data update through Procedure
- 83. DDL-23.** Example of Data Delete through Procedure.
- 84. DML-61.** Justification of Data Delete through Procedure
- 85. DDL-24.** Example of Using Output Parameter.
- 86. DML-62.** Justification of OUTPUT PROCEDURE
- 87. DDL-25.** Alter Procedure.
- 88. DML-63.** Justification of ALTER PROCEDURE
- 89. DDL-26.** Example of Procedure with error handle.
- 90. DML-64.** Justification of Procedure with error handle
- 91. DDL-27.** Example of Data Modification through Single Procedure-
(select,Insert,Update,Delete,Output,Return)
- 92. DML-65.** Justification of Data Modification and output return through Single
Procedure(Insert,Update,DELETE,OUTPUT,RETURN)
- 93. DDL-28.** DELETE PROCEDURE.
- 94. DDL-29.** Example of creating Index.
- 95. DML-66.** Justification of index creation
- 96. DDL-30.** Example of deleting Index.
- 97. DML-67.** Justification of index deletion
- 98. DDL-31.** Creating After Trigger.
- 99. DML-68.** Justification of TRIGGER(For/After trigger)
- 100. DDL-32.** Creating Instead of Trigger.
- 101. DML-69.** Justification of Instead Of trigger
- 102. DDL-33.** DELETE TRIGGER.
- 103. DDL-34.** Example of Creating Sequence.
- 104. DDL-35.** Example of Sequence Modification (Alter Sequence).

- 105. DDL-36.** Example of Truncate table.
- 106. DML-70.** Justification of truncate Table
- 107. DDL-37.** Create Log In.
- 108. DDL-38.** Log in Modification (ALTER LOG IN).
- 109. DDL-39.** Delete Log in.
- 110. DDL-40.** Create user.
- 111. DDL-41.** User Modification (Alter User)
- 112. DDL-42.** Delete User.
- 113. DDL-43.** Example of Giving Permission.
- 114. DDL-44.** Example of Permission Taking out.